

SECURE CODING REVIEW

IDENTIFYING AND PREVENTING
SECURITY VULNERABILITIES

CYBER SECURITY PRESENTATION
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INTERNSHIP PROGRAM

INTRODUCTION

Insecure coding practices can expose software applications to cyber threats, unauthorized access, and data breaches. Secure coding helps developers identify vulnerabilities and implement security best practices to build safer applications.

This project focuses on reviewing vulnerable code examples and implementing secure coding techniques using Python.

WHAT IS SECURE CODING?

Secure coding is the practice of developing software in a way that minimizes vulnerabilities and protects applications from cyber threats.

Objectives

- ✓ Prevent security vulnerabilities
- ✓ Protect sensitive data
- ✓ Reduce cyber risks
- ✓ Improve application security



COMMON SECURITY VULNERABILITIES



- 🔒 Hardcoded Credentials
- ⚠️ Weak Password Handling
- 💉 SQL Injection
- 📥 Improper Input Validation
- 📁 Unsafe File Handling

Impact:

These vulnerabilities may lead to unauthorized access, data theft, and application compromise.

VULNERABLE CODE EXAMPLE



CODE SNIPPET

```
if username == "admin" and  
password == "admin123":  
    print("Login Successful")
```

Security Issues

- ✗ Hardcoded credentials
- ✗ Weak authentication method
- ✗ Easy to guess passwords

Risk

Attackers can gain unauthorized access easily.

VULNERABLE_LOGIN.PY RUNNING OUTPUT

```
PROBLEMS  OUTPUT  TERMINAL  ...  
  
PS C:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview> & 'C:\Program Files\Python314\python.exe' 'c:\Users\Niharika Badavath\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '58701' '--' 'c:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview\vulnerable_login.py'  
Enter username: niharika  
Enter password: Niharika@123  
Login Successful  
PS C:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview>
```

RISKS OF INSECURE CODING

- ⚠ Unauthorized system access
- ⚠ Data breaches
- ⚠ Password theft
- ⚠ Information leakage
- ⚠ Financial and reputational damage

EXAMPLE

Weak authentication systems make applications vulnerable to cyberattacks.



SECURE CODE EXAMPLE

CODE SNIPPET

```
if username == stored_username and password  
== stored_password:  
print("Secure Login Successful")
```

Security Improvements:

- ✓ Improved authentication
- ✓ Better password handling
- ✓ More secure coding structure

Benefit:

Secure coding reduces vulnerabilities
and improves application security.

SECURE_LOGIN.PY OUTPUT

```
PROBLEMS  OUTPUT  TERMINAL  ...  
  
PS C:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview> & 'C:\Program Files\Python314\python.exe' 'c:\Users\Niharika Badavath\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '55126' '--' 'c:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview\secure_login.py'  
Enter username: vinny  
Enter password: 12345  
Invalid Credentials  
PS C:\Users\Niharika Badavath\OneDrive\Desktop\CodeAlpha_SecureCodingReview> 
```

SECURE CODING BEST PRACTICES

- ✓ Use strong passwords
- ✓ Avoid hardcoded credentials
- ✓ Validate user inputs
- ✓ Encrypt sensitive data
- ✓ Perform regular security testing
- ✓ Keep software updated

Important Note :
Security should be integrated throughout the
software development lifecycle.



KEY LEARNING OUTCOMES

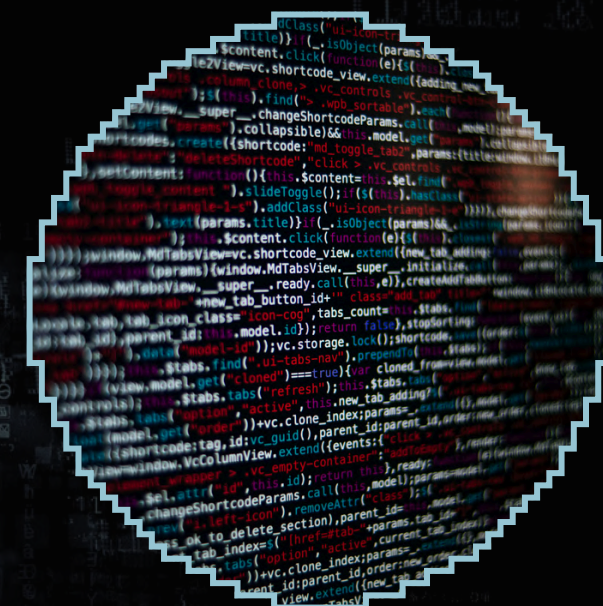
- ✓ Importance of secure coding
- ✓ Identification of vulnerabilities
- ✓ Risks of insecure applications
- ✓ Implementation of secure coding practices
- ✓ Importance of cybersecurity in software development

CONCLUSION

Secure coding practices play a vital role in protecting software applications from cyber threats and unauthorized access. Identifying vulnerabilities and implementing secure coding techniques helps build safer and more reliable systems.

FINAL MESSAGE

"Secure Code is the Foundation of Secure Systems."





THANK YOU

“Security is not a feature, it is a responsibility.”

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